DATA CENTER OPERATIONS

SYSTEMWIDE

Audit Report 12-30
April 23, 2013

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ABBREVIATIONS

ATS	Academic Technology Services
CIO	Chief Information Officer
CO	Office of the Chancellor
CSU	California State University
F&A	Finance and Administration
FISMA	Financial Integrity and State Manager's Accountability Act
ICSUAM	Integrated California State University Administrative Manual
I&IT	Instructional and Information Technology
IT	Information Technology
ITS	Information Technology Services
MDF	Main Distribution Frame
MIS	Management Information Systems
OUA	Office of the University Auditor
SAM	State Administrative Manual
T&C	Technology and Communication

EXECUTIVE SUMMARY

As a result of a systemwide risk assessment conducted by the Office of the University Auditor (OUA) during the last quarter of 2011, the Board of Trustees, at its January 2012 meeting, directed that *Data Center Operations* be reviewed. The OUA had previously reviewed some aspects of *Data Center Operations* in the 2008 and 2009 audits of *Information Security* and the 2010 and 2011 audits of *IT Disaster Recovery Planning*. The OUA also reviewed *Data Center Operations* in the biennial Financial Integrity and State Manager's Accountability Act (FISMA) audits, the last of which was performed on campuses in 2009.

We visited six campuses from February 20, 2012, through September 7, 2012, and audited the procedures in effect at that time. Campus-specific findings and recommendations have been discussed and reported individually.

In our opinion, due to the effect of the weaknesses described below, the fiscal, operational and administrative controls for data center operations as of September 7, 2012, taken as a whole, were not sufficient to meet the objectives stated in the "Purpose" section of this report. Areas of major concern include: physical security, fire protection and environmental controls, and emergency preparedness and training.

As a result of changing conditions and the degree of compliance with procedures, the effectiveness of controls changes over time. Specific limitations that may hinder the effectiveness of an otherwise adequate system of controls include, but are not limited to, resource constraints, faulty judgments, unintentional errors, circumvention by collusion, and management overrides. Establishing controls that would prevent all these limitations would not be cost-effective; moreover, an audit may not always detect these limitations.

Our audit did not examine all controls over data center operations but was designed to assess management controls, increase awareness of the topic, and assess regulatory compliance for significant data center operations categories that are prevalent in the California State University environment.

The following summary provides management with an overview of conditions requiring attention. Areas of review not mentioned in this section were found to be satisfactory. Numbers in brackets [ ] refer to page numbers in the report.

PHYSICAL SECURITY [7]

Background checks were not performed on all employees who had physical access to the campus data centers or systems that contain sensitive data. In addition, access to campus data center rooms was not always limited to those individuals whose responsibilities required access. Further, restriction of physical access to data rooms and monitoring of personnel who entered and exited those rooms needed improvement. For example, the doors to campus data centers and network main distribution frame rooms were not always equipped with electronic access control systems to track and monitor personnel entering and exiting the facilities, and most campuses did not have procedures to regularly monitor access reports from existing electronic key card systems. Also, data and computing facilities were not always adequately protected to detect unauthorized physical access.

FIRE PROTECTION AND ENVIRONMENTAL CONTROLS [11]

The campuses had not provided adequate fire detection and suppression controls to ensure that facilities were protected, and satellite data centers operated by campus departments did not always have adequate fire suppression or environmental monitoring systems.

EMERGENCY PREPAREDNESS AND TRAINING [13]

Emergency power generators capable of sustaining computer operations in the event of a power outage were not in use at all campus data centers. In addition, data center emergency preparedness and training required improvement. Specifically, campuses did not always have documented data center emergency operations procedures, and data center personnel were not always trained in the operation of the fire suppression system and the use of fire extinguishers.

INTRODUCTION

BACKGROUND

Integrated California State University Administrative Manual (ICSUAM) §8000.0, *Information Security Policy*, dated April 19, 2010, represents the most recent and specific guidance to campuses regarding the security and protection of data center operations. It provides direction for managing and protecting the confidentiality, integrity, and availability of California State University (CSU) information assets and defines the organizational scope of information security throughout the system. Specifically, the policy states that the Board of Trustees is responsible for protecting the confidentiality, integrity, and availability of CSU information assets. Unauthorized modification, deletion, or disclosure of information assets can compromise the mission of the CSU, violate individual privacy rights, and possibly constitute a criminal act.

ICSUAM §8000.0 further states that it is the collective responsibility of all users to ensure the confidentiality of information that the CSU must protect from unauthorized access; the integrity and availability of information stored on or processed by CSU information systems; and compliance with applicable laws, regulations, and CSU or campus policies governing information security and privacy protection.

The policy applies to all campuses; central and departmentally managed campus information assets; all users employed by campuses or any other person with access to campus information assets; all categories of information, regardless of the medium in which the information asset is held or transmitted (e.g., physical or electronic); and information technology facilities, applications, hardware systems, and network resources owned or managed by the CSU.

ICSUAM §8080 states that each campus must identify physical areas that must be protected from unauthorized physical access. Such areas include data centers and other locations on the campus where information assets containing protected data are stored. Campuses must protect these limited-access areas from unauthorized physical access while ensuring that authorized users have appropriate access. Campus information assets that access protected data located in public and non-public access areas must be physically secured to prevent theft, tampering, or damage. The level of protection provided must be commensurate with that of identifiable risks. Campuses must review and document physical access rights to campus limited-access areas annually.

State Administrative Manual (SAM) §5330 states that physical security practices prevent unauthorized physical access, damage, and interruption to an agency's assets. Physical security practices for each facility must be adequate to protect the most sensitive information technology application housed in that facility. Agencies must take the appropriate physical security measures to provide for: management control of physical access to information assets (including personal computer systems, computer terminals, and mobile devices) by agency staff and outsiders; prevention, detection, and suppression of fires; and prevention, detection, and minimization of water damage and loss or disruption of operational capabilities due to electrical power fluctuations or failure.

SAM §5335 states that agencies are responsible for the management and operation of their information processing facilities. The security program should identify and document the appropriate practices to

ensure the integrity and security of the agency's information assets. SAM §5335 references International Standards Organization 17799 Section 9, Physical and Environmental Security, and National Institute of Standards and Technology Special Publication 800-12 (Chapter 15), along with other standards and guidance criteria.

Historically, data center operations were reviewed by the CSU Office of the University Auditor (OUA) as part of cyclical audits based on the Financial Integrity and State Manager's Accountability Act (FISMA) of 1983, passed by the California Legislature and detailed in Government Code §13400 through §13407. Beginning in calendar year 2010, cyclical FISMA audits were reevaluated and discontinued due to a change in the OUA audit risk assessment methodology. Using the new procedure, the OUA worked with CSU campus executive management to identify high-risk areas on each campus. Data Center Operations was selected as a high-risk area to review in 2012.

PURPOSE

Our overall audit objective was to ascertain the effectiveness of existing policies and procedures related to the administration and control of data center operations; determine the adequacy of controls over the related processes; and ensure compliance with relevant governmental regulations, Trustee policy, Office of the Chancellor directives, and campus procedures.

Within the overall audit objective, specific goals included determining whether:

- Certain essential administrative and managerial internal controls are in place, including delegations of authority and responsibility, management committees, and documented policies and procedures.
- Data processing facilities employ physical security safeguards for achieving and maintaining appropriate protection of organizational assets.
- Data processing facilities contain adequate fire supression provisions and employ controls that help maintain a proper operating environment.
- Handling procedures for backup media ensure that the movement and storage of tapes is controlled and accountable.
- Formal event reporting and escalation procedures are in place for job scheduling.
- Change management procedures are sufficient to ensure that modifications to the systems or network are authorized.
- Management review of help desk activities ensures a proactive approach toward determining whether there is a systemic cause to problems reported.

SCOPE AND METHODOLOGY

The proposed scope of the audit as presented in Attachment A, Audit Agenda Item 2 of the January 24 and 25, 2012, meeting of the Committee on Audit stated that *Data Center Operations* would include review and compliance with Trustee policy, federal and state directives, and campus policies and procedures; physical security provisions; environmental controls; processing and scheduling controls; backup and recovery processes; and emergency preparations.

Our study and evaluation were conducted in accordance with the *International Standards for the Professional Practice of Internal Auditing* issued by the Institute of Internal Auditors and included the audit tests we considered necessary in determining that operational and administrative controls are in place and operative. This review emphasized, but was not limited to, compliance with state and federal laws, Board of Trustee policies, and Office of the Chancellor and campus policies, letters, and directives. The audit review focused on procedures currently in effect.

We focused primarily upon the administrative, compliance, operational, and technical controls over the campus data center, network rooms, and personnel operations. Specifically, we reviewed and tested:

- Data center policies and procedures.
- Computer operations organizational structure and management framework.
- Physical security over data processing facilities.
- Fire prevention and environmental controls.
- Emergency preparedness and training.
- Storage and handling of backup media.
- Job scheduling.
- Change management.
- Help desk support.

Our testing and methodology was designed to provide a managerial-level review of key data processing practices over data center operations. Our review did not examine all categories of computer operations; selected IT processes not related to the data center or related data processing facilities were excluded from the scope of the review. Our testing approach was designed to provide a view of the security and controls used to protect only key computing and business processes.

During the course of the audit, we visited six campuses: Channel Islands, Chico, Dominguez Hills, Long Beach, East Bay, and Pomona. We interviewed campus personnel and audited procedures in effect at the time of the audit.

OBSERVATIONS, RECOMMENDATIONS, AND MANAGEMENT RESPONSES

PHYSICAL SECURITY

BACKGROUND CHECKS

Background checks were not performed on all employees who had physical access to the campus data centers or systems that contain sensitive data.

Integrated California State University Administrative Manual (ICSUAM) §8030, *Personnel Information Security*, dated April 19, 2010, states that campuses must develop procedures to conduct background checks on positions involving access to level one information assets as defined in the California State University (CSU) Data Classification Standard.

Campus management stated various reasons for background checks not being performed, including lack of awareness that checks were required and the understanding that the Office of the Chancellor (CO) was developing a systemwide policy on background checks to address compliance with applicable laws, policies, and bargaining unit agreements, which the campuses planned to adopt.

Failure to perform background checks on personnel who have access to sensitive data increases the risk of potential mishandling and inappropriate disclosure of sensitive data.

Recommendation 1

We recommend that the CO reiterate to the campuses that background checks are required for all employees who have physical access to the campus data centers or systems that contain sensitive data.

Management Response

Management concurs. The CO will remind the campuses that background checks are required per ICSUAM §8030, and that background checks should be conducted in compliance with CSU Human Resources and Labor Relations policy. This notification will take place by September 30, 2013.

DATA CENTER ACCESS

Access to campus data center rooms was not always limited to those individuals whose responsibilities required access.

Specifically, we noted that:

- Several individuals were incorrectly granted access, and others with access no longer required it or had been terminated.

- Individuals from various campus departments, such as custodial, facilities, and police services, had been granted data center access, even though their current responsibilities did not warrant it. Most individuals with questionable access were given master keys rather than electronic key cards that would have allowed management to track and monitor when they entered and exited the room.

ICSUAM §8080, *Physical Security*, dated April 19, 2010, states that each campus must identify physical areas that must be protected from unauthorized physical access. Such areas would include data centers and other locations on the campus where information assets containing protected data are stored. Campuses must protect these limited-access areas from unauthorized physical access while ensuring that authorized users have appropriate access. Campus information assets that access protected data that are located in public and non-public access areas must be physically secured to prevent theft, tampering, or damage. The level of protection provided must be commensurate with that of identifiable risks. Campuses must review and document physical access rights to campus limited-access areas annually.

Campus management stated various reasons for the inappropriate physical access and the lack of policies and procedures over monitoring, including a lack of guidance on the access roles of campus support personnel and failure to properly consider access via physical master keys.

Inadequate physical security over information technology assets and sensitive data information increases the risk that unauthorized personnel will have access to information assets.

Recommendation 2

We recommend that the CO:

a. Ensure that campuses limit access to campus data center rooms to those individuals whose responsibilities require access.

b. Remind the campuses that they are required to periodically review and validate physical access rights to server rooms and data centers housing systems containing level 1 sensitive data.

c. Ensure that the campus CIOs submit an annual certification of compliance with physical security standards and procedures to include an attestation that physical security rights have been reviewed and are appropriate.

Management Response

Management concurs. The CO will remind the campuses of their obligations under ICSUAM §8080. The CO will also notify the campuses that they are required to submit an annual certification that they are in compliance with the related physical security standards and procedures. This notification will take place not later than September 30, 2013.

FACILITY ACCESS CONTROL AND MONITORING

Restriction of physical access to data rooms and monitoring of personnel who entered and exited those rooms needed improvement.

Specifically, we found that:

- The doors to campus data centers and network main distribution frame (MDF) rooms housing systems with level 1 data were not always equipped with electronic access control systems to track and monitor personnel entering and exiting the facilities.

- Most campuses did not have procedures to regularly monitor access reports from existing electronic key card systems.

ICSUAM §8080, *Physical Security*, dated April 19, 2010, states that each campus must identify physical areas that must be protected from unauthorized physical access. Such areas would include data centers and other locations on the campus where information assets containing protected data are stored.

State Administrative Manual (SAM) §5330 states that physical practices for each facility must be adequate to protect the most sensitive information technology application housed in that facility. Agencies must take appropriate physical security measures to provide for control of physical access to information assets by agency staff and outsiders.

Campus management stated that electronic locks had not been placed in all facilities due to the high cost of installation. Management further stated that they were unaware of the need to monitor door access logs, and that the MDFs were not routinely inspected due to oversight.

Failure to adequately restrict access and monitor personnel who enter and exit the data center and adequately secure network devices could increase the risk of unauthorized or inappropriate activity occurring without detection.

Recommendation 3

We recommend that the CO ensure that campuses:

a. Equip data centers and MDF rooms housing systems with level 1 data with electronic access control systems to track and monitor personnel entering and exiting the facilities.

b. Establish procedures to regularly monitor access reports from electronic key card systems.

c. Ensure that the campus CIOs submit an annual certification that physical security controls are equipped on all data centers housing level 1 data.

Management Response

Management concurs. The CO will remind the campuses of their obligations under ICSUAM §8080 and §5330.

Campuses will be advised to: a) equip data centers and MDF rooms housing systems with level 1 data with electronic access control systems, or relocate systems with level 1 data to facilities that are so equipped; and b) establish procedures to regularly monitor access reports from the access control systems.

The CO will also notify the campuses that they are required to submit an annual certification that they are in compliance with the related physical security standards and procedures. This notification will take place not later than September 30, 2013.

ALARM SYSTEMS

Data and computing facilities were not always adequately protected to detect unauthorized physical access.

Specifically, we noted that unattended data centers, server rooms, and network MDF rooms housing systems with level 1 data did not always have security alarm systems installed to detect entry.

ICSUAM §8080, *Physical Security*, dated April 19, 2010, states that each campus must identify physical areas that must be protected from unauthorized physical access. Such areas would include data centers and other locations on the campus where information assets containing protected data are stored. Campuses must protect these limited-access areas from unauthorized physical access while ensuring that authorized users have appropriate access. Campus information assets that access protected data that are located in public and non-public access areas must be physically secured to prevent theft, tampering, or damage. The level of protection provided must be commensurate with that of identifiable risks. Campuses must review and document physical access rights to campus limited-access areas annually.

SAM §5330 states that physical security practices for each facility must be adequate to protect the most sensitive information technology application housed in that facility. Agencies must take appropriate physical security measures to provide for control of physical access to information assets by agency staff and outsiders.

The campus chief information officers (CIO) stated that alarms had not been installed in data centers because the rooms were located in buildings that were secured after hours. They further stated that security alarms had not been placed in the MDF rooms due to installation and monitoring costs.

Failure to equip computer rooms with security alarm systems increases the risk of security breaches and theft of computing equipment.

Recommendation 4

We recommend that the CO:

a. Remind the campuses that they are required to identify and protect data centers, server rooms, MDF rooms, and any other area housing systems containing level 1 or level 2 sensitive data.

b. Ensure that the campus CIOs submits an annual certification of compliance with systemwide physical security policies.

Management Response

Management concurs. The CO will remind the campuses of their obligations under ICSUAM §8080 and §5330. The CO will also notify the campuses that they are required to submit an annual certification that they are in compliance with systemwide physical security policies. This notification will take place not later than September 30, 2013.

FIRE PROTECTION AND ENVIRONMENTAL CONTROLS

FIRE PROTECTION

The campuses had not provided adequate fire detection and suppression controls to ensure that facilities were protected.

Specifically, we noted that campus primary data centers did not always have smoke detection systems or automatic fire suppression systems.

SAM §5330 states that physical practices for each facility must be adequate to protect the most sensitive information technology application housed in that facility. Agencies must take appropriate physical security measures to provide for prevention, detection, and suppression of fires.

The campus CIOs stated their belief that passing the building fire inspection was confirmation that their data center fire provisions were sufficient.

A lack of smoke detectors and fire suppression equipment in data centers increases the risk that information assets will be damaged during disasters or emergencies.

Recommendation 5

We recommend that the CO validate that the campuses have implemented adequate fire detection and suppression controls to protect primary data centers in the event of fire.

Management Response

Management concurs. The CO will remind the campuses of their obligations under ICSUAM §5330. The CO will also ensure that campuses submit an annual certification that they are in compliance with the related physical security and fire safety standards and procedures. This notification will take place not later than September 30, 2013.

SATELLITE DATA CENTER CONTROLS

Satellite data centers operated by campus departments did not always have adequate fire suppression or environmental monitoring systems.

We reviewed several departmental data centers, and we found that they did not always have automatic fire suppression systems, fire extinguishers, smoke detectors, water sensors, and temperature and environmental monitoring devices.

SAM §5330 states that physical security practices for each facility must be adequate to protect the most sensitive information technology application housed in that facility. Agencies must take appropriate physical security measures to provide for prevention, detection, and suppression of fires.

The campus CIOs stated that fire protection measures and environmental controls were not consistently in place because data center management was highly decentralized, and standards regarding what constitutes adequate fire suppression for a data center had not been developed.

A lack of automatic fire suppression systems, fire extinguishers, smoke detectors, and environmental controls in data centers increases the risk that information assets will be damaged during disasters or emergencies.

Recommendation 6

We recommend that the CO ensure that campuses have provided adequate fire suppression and environmental monitoring systems in satellite data centers, or ensure that the campuses consolidate those facilities into data centers that provide such controls.

Management Response

Management concurs. The CO will remind the campuses of their obligations under ICSUAM §5330. Campuses will be advised to equip satellite data centers housing systems with level 1 data with appropriate fire suppression and environmental monitoring systems, or relocate systems with level 1 data to facilities that provide such controls. This notification will take place not later than September 30, 2013.

EMERGENCY PREPAREDNESS AND TRAINING

EMERGENCY POWER

Emergency power generators capable of sustaining computer operations in the event of a power outage were not in use at all campus data centers.

ICSUAM §8085, *Business Continuity and Disaster Recovery*, dated April 19, 2010, states that each campus must ensure that information assets can, in case of a catastrophic event, continue to operate and be appropriately accessible to users.

SAM §5330 states that physical security practices for each facility must be adequate to protect the most sensitive information technology application housed in that facility. Agencies must take appropriate physical security measures to provide for prevention, detection, and minimization of loss or disruption of operational capabilities due to electrical power fluctuations or failure.

The campus CIOs stated that backup power generators had not been purchased or installed due to the extremely high cost.

A lack of backup generators capable of sustaining computer operations increases the risk that the campus could lose the ability to provide data processing services in the event of a power outage, which could disrupt campus operations.

Recommendation 7

We recommend that the CO ensure that campuses install emergency power generators capable of sustaining computer operations in the event of a power outage in data center rooms, or move equipment into computing facilities that have backup power.

Management Response

Management concurs. The CO will remind the campuses of their obligations under ICSUAM §8085 and §5330. Campuses will be advised to install emergency power generators in satellite data centers housing systems with level 1 data, or relocate systems with level 1 data to facilities with such equipment. This notification will take place not later than September 30, 2013.

PROCEDURES AND TRAINING

Data center emergency preparedness and training required improvement.

We found that:

- Campuses did not always have documented data center emergency operations procedures.

- Data center personnel were not always trained in the operation of the fire suppression system and the use of fire extinguishers.

SAM §5355.1 states that disaster recovery plans and other IT procedures should be developed to ensure that critical services and applications are restored as quickly as possible and with minimal loss of data.

The campus CIOs stated that the lack of formal data center emergency operations procedures and the failure to provide fire suppression and safety equipment training to data center personnel were due to oversight.

Inadequate data center emergency systems procedures and training could lead to misunderstandings and confusion in the event of a disaster or emergency.

Recommendation 8

We recommend that the CO:

a. Ensure that campuses develop and test data center emergency operations procedures.

b. Issue specific guidance regarding requirements for training data center personnel on fire suppression and safety equipment.

Management Response

Management concurs. The CO will remind the campuses of their obligations under ICSUAM §5355.1. Campuses will be advised to develop and test data center emergency operations procedures, paying particular attention to fire and life safety. This notification will take place not later than September 30, 2013.

APPENDIX A:
PERSONNEL CONTACTED

Name	Title
Chancellor's Office	
Benjamin F. Quillian	Executive Vice Chancellor and Chief Financial Officer
Bruce Briggs	Chief Information Officer
Mark Crase	Senior Director, Technology Infrastructure Services
William Perry	Chief Information Security Officer
Michael Redmond	Senior Director, Chancellor's Office Fiscal Control and Special Initiatives
California State University, Channel Islands	
Richard R. Rush	President
Pamela Abbot-Mouchou	Executive Administrative Assistant, Finance and Administration (F&A)
Herbert Aquino	Manager of Information Technology Infrastructure, Technology and Communication (T&C)
Melissa Bergem	Project Coordinator, T&C
Michael Berman	Vice President, T&C
Will Brogdon	Student, T&C
Caroline Doll	Director, F&A
Diana Enos	Human Resources Manager, F&A
Liza Ernst	Programs Assistant, Human Resources
Neal Fisch	Director Application Services, T&C
Judy Frazier	Budget Analyst, T&C
Ernesto Gutierrez	Senior Network Analyst, T&C
Theresa Kocis	Fire Alarm Specialist, Operations
Kristin Steiner	Administrative Support Coordinator, T&C
Aaron Lasely	Supervising Locksmith, Operations
Michael Long	Senior Telecommunications Analyst, T&C
Nasser Mansour	Database Administrator, T&C
Peter Mosinskis	Director of Information Technology Strategy, T&C
Jess Paredes	Operating Systems Analyst, T&C
Richard Paulson	Supervising Building Service Engineer, Operations
Anna Pavin	Associate Vice President, Human Resources
Ysabel Trinidad	Vice President, F&A
Indy Valencia	Manager User Services, T&C
California State University, Chico	
Paul J. Zingg	President
Sharyn Abernatha	Assistant Vice President for Staff Human Resources
Scott Claverie	Director of Communications Services
Lee Cummings	Supervisor, Telecommunication Services
Amy Gremore	Manager of Employment Services

California State University, Chico (cont.)

Lorraine Hoffman	Vice President for Business and Finance
Ed Hudson	Information Security Officer
Jeni Kitchell	Interim Director of University Budget and Resource Management
Andrea Mox	Manager, Information Technology (IT) Support Services
Mike Murray	Administrative Desktop Management
Dennis Partington	Network Manager
Ray Quinto	Manager, Enterprise Systems
Ryan Richter	Labs and Academic Desktop Management
Jerry Ringel	Director, Computing and Technology Support
Tom Rosenow	Director, Application Development and Technical Support, Enrollment Management
Michael Schilling	Vice Provost for Information Resources and Chief Information Officer
Kevin Weherly	Desktop Support Services

California State University, Dominguez Hills

Willie Hagan	President
Mildred Garcia	President (At time of review)
Mahabub Alam	Network Specialist, Campus Backbone Network
Ron Bergmann	Associate Vice President and Chief Information Officer, IT
Ed Laio	Network Analyst
Danny Lujan	Director of Infrastructure
Mary Ann Rodriquez	Vice President, Administration and Finance
Jonathan Scheffler	Associate Director, Physical Plant
Karen Wall	Associate Vice President, Administration and Finance
Muneca Williams	IT Administrative Support Assistant

California State University, East Bay

Leroy M. Morishita	President
Rich Avila	Director, Server and Network Operations
Lee Breitzman	Lead Operations Specialist, Server Operations
Matt Collins	Director, Application Systems
Chris Da Silva	Network Analyst
Maggie Graney	Director, Compliance and Internal Control
Jim Hodges	Chief of Police
Audrey Katzman	Information Technology Business Manager
Gene Lim	Director, Server Operations Services
David Miller	Maintenance Manager, Facilities Management
Setareh Sarrafan	Director of User Support Services
Borre Ulrichsen	Deputy Chief Information Officer
Brad Wells	Vice President and Chief Financial Officer, Administration and Finance

California State University, Long Beach

Donald J. Para	Interim President
F. King Alexander	President (At time of review)
Martha Carey	Data Center Operator
Janet Foster	Associate Vice President, Information Technology Services (ITS)
Vickie Hamilton	Director, Human Resources
Byron Jackson	Director of Service Management and Operations, ITS
Craig Kleen	Assistant Director of Network Services, ITS
Roman Kochan	Interim Associate Vice President of Academic Technology Services (ATS) and Dean and Director, University Library
Steve La	Director of Network, Telecommunication and Security, ITS
Michael Markoski	Director of Data Center and Emergency Management, ITS
Greg Pascal	Communications/Information System Manager, University Police
Cheryl Perkins	Administrative Services Manager
Wayne Pierson	Director of Servers, Systems and Websites, ITS
Aysu Spruill	Director of Internal Auditing/Information Security Officer
Mary Stephens	Vice President, Administration and Finance
Ryan Tapp	Library, Desktop and User Support, ATS
Don Taylor	Operating Systems Technical Lead, ITS Servers, Systems and Web

California State Polytechnic University, Pomona

J. Michael Ortiz	President
Albert Arboleda	Information Security Officer, Instructional and Information Technology (I&IT) Information Security
Mark Bailey	Department Engineer, Electrical and Computer Engineering Department
Edwin Barnes III	Vice President, Administrative Affairs
Cathy Bates	Director, Student Affairs Applications and Technology Solutions/ Student Affairs Information and Technology Services
Jarod Beekman	Associate Director of Networking and Telecommunications, I&IT Systems
Eric Bellman	Manager Systems Operations, Administrative Affairs Information Systems
Joseph Bustamante	Information Systems Analyst, Management Information Systems (MIS) Foundation
David Drivdahl	Interim Director of System Administration, I&IT Systems
Joseph Galdonik	Systems Support Technician, College of Engineering
Kathy Harper	Assistant to the Vice President, Finance and Administrative Services
Hovig Krikorian	Associate Systems Analyst, College of the Extended University
Gabriel Kuri	Senior Network Engineer, I&IT Systems
Christopher Laasch	Information Technology Administrative, Student Affairs Information and Technology Services
Darwin Labordo	Associate Vice President, Finance and Administrative Services
Cecilia Lenasdotter	Information Systems Analyst, MIS Foundation
Whitney Lopez	Administrative Support Coordinator, I&IT Systems
Walter Marquez	Director, Facilities Administrative and Energy Services

California State Polytechnic University, Pomona (cont.)

Mary Martinez	Payroll Manager, Human Resource Services
Debbi McFall	Emergency Service Coordinator, University Police
John McGuthry	Chief Information Officer, I&IT
Mark Miller	Director, Facilities Management and Capital Projects
Kevin Morningstar	Executive Director, Student Affairs Information and Technology Services
Denton Mosier	Executive Director Technical Support, I&IT Support
Son Phan	Senior Systems Analyst, College of Engineering
Sharon Reiter	Associate Vice President, Human Resource Services
James Schneider	Network Analyst, I&IT Systems
Randall Townsend	Management Information Systems Manager, MIS Foundation
Mark VanDusen	Systems Support Technician, College of Engineering
Joice Xiong	Director of Internal Audits, Administrative Affairs
Glendy Yeh	Executive Director, Administrative Affairs Information Systems

The California State University
OFFICE OF THE CHANCELLOR

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MEMORANDUM

RECEIVED
UNIVERSITY AUDITOR
JUL 19 2013
THE CALIFORNIA STATE UNIVERSITY

DATE: July 19, 2013

TO: Larry Mandel
University Auditor

FROM: Benjamin F. Quillian
Executive Vice Chancellor and
Chief Financial Officer

SUBJECT: **Audit Report # 12-30 Data Center Operations Audit**

In response to the "Incomplete Draft" report dated April 23, 2013, we are providing the enclosed management responses.

Should you have any questions, please feel free to contact us.

BFQ:mpr

Attachment

cc: Bruce Briggs, AVC/CIO InformationTechnology
Mark Crase, Chief Technology Officer
William Perry, Chief Info Management & Security Officer
Michael Redmond, Sr. Dir. CO Budget and Special Initiatives

DATA CENTER OPERATIONS

SYSTEMWIDE

Audit Report 12-30

PHYSICAL SECURITY

BACKGROUND CHECKS

Recommendation 1

We recommend that the CO reiterate to the campuses that background checks are required for all employees who have physical access to the campus data centers or systems that contain sensitive data.

Management Response

Management concurs. The CO will remind the campuses that background checks are required per ICSUAM §8030, and that background checks should be conducted in compliance with CSU Human Resources and Labor Relations policy. This notification will take place by September 30, 2013.

DATA CENTER ACCESS

Recommendation 2

We recommend that the CO:

a. Ensure that campuses limit access to campus data center rooms to those individuals whose responsibilities require access.

b. Remind the campuses that they are required to periodically review and validate physical access rights to server rooms and data centers housing systems containing level 1 sensitive data.

c. Ensure that the campus CIOs submit an annual certification of compliance with physical security standards and procedures to include an attestation that physical security rights have been reviewed and are appropriate.

Management Response

Management concurs. The CO will remind the campuses of their obligations under ICSUAM §8080. The CO will also notify the campuses that they are required to submit an annual certification that they are in compliance with the related physical security standards and procedures. This notification will take place not later than September 30, 2013.

FACILITY ACCESS CONTROL AND MONITORING

Recommendation 3

We recommend that the CO ensure that campuses:

a. Equip data centers and MDF rooms housing systems with level 1 data with electronic access control systems to track and monitor personnel entering and exiting the facilities.

b. Establish procedures to regularly monitor access reports from electronic key card systems.

c. Ensure that the campus CIOs submit an annual certification that physical security controls are equipped on all data centers housing level 1 data.

Management Response

Management concurs. The CO will remind the campuses of their obligations under ICSUAM §8080 and §5330.

Campuses will be advised to: a) equip data centers and MDF rooms housing systems with level 1 data, with electronic access control systems or to relocate systems with Level 1 data to facilities that are so equipped; and b) establish procedures to regularly monitor access reports from the access control systems.

The CO will also notify the campuses that they are required to submit an annual certification that they are in compliance with the related physical security standards and procedures. This notification will take place not later than September 30, 2013.

ALARM SYSTEMS

Recommendation 4

We recommend that the CO:

a. Remind the campuses that they are required to identify and protect data centers, server rooms, MDF rooms, and any other area housing systems containing level 1 or level 2 sensitive data.

b. Ensure that the campus CIOs submits an annual certification of compliance with system-wide physical security policies.

Management Response

Management concurs. The CO will remind the campuses of their obligations under ICSUAM §8080 and §5330. The CO will also notify the campuses that they are required to submit an annual certification that they are in compliance with system-wide physical security policies. This notification will take place not later than September 30, 2013.

FIRE PROTECTION AND ENVIRONMENTAL CONTROLS

FIRE PROTECTION

Recommendation 5

We recommend that the CO validate that the campuses have implemented adequate fire detection and suppression controls to protect primary data centers in the event of fire.

Management Response

Management concurs. The CO will remind the campuses of their obligations under ICSUAM §5330. The CO will also ensure that campuses submit an annual certification that they are in compliance with the related physical security and fire safety standards and procedures. This notification will take place not later than September 30, 2013.

SATELLITE DATA CENTER CONTROLS

Recommendation 6

We recommend that the CO ensure that campuses have provided adequate fire suppression and environmental monitoring systems in satellite data centers, or ensure that the campuses consolidate those facilities into data centers that provide such controls.

Management Response

Management concurs. The CO will remind the campuses of their obligations under ICSUAM §5330. Campuses will be advised to equip satellite data centers housing systems with level 1 data, with appropriate fire suppression and environmental monitoring systems, or to relocate systems with Level 1 data to facilities that provide such controls. This notification will take place not later than September 30, 2013.

EMERGENCY PREPAREDNESS AND TRAINING

EMERGENCY POWER

Recommendation 7

We recommend that the CO ensure that campuses install emergency power generators capable of sustaining computer operations in the event of a power outage in data center rooms, or move equipment into computing facilities that have backup power.

Management Response

Management concurs. The CO will remind the campuses of their obligations under ICSUAM §8085 and §5330. Campuses will be advised to install emergency power generators in satellite data centers housing systems with level 1 data, or to relocate systems with Level 1 data to facilities with such equipment. This notification will take place not later than September 30, 2013.

PROCEDURES AND TRAINING

Recommendation 8

We recommend that the CO:

a. Ensure that campuses develop and test data center emergency operations procedures.

b. Issue specific guidance regarding requirements for training data center personnel on fire suppression and safety equipment.

Management Response

Management concurs. The CO will remind the campuses of their obligations under ICSUAM §5355.1. Campuses will be advised to develop and test data center emergency operations procedures, paying particular attention to fire and life safety. This notification will take place not later than September 30, 2013.

THE CALIFORNIA STATE UNIVERSITY
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September 4, 2013

MEMORANDUM

TO: Mr. Larry Mandel
University Auditor

FROM: Timothy P. White
Chancellor

SUBJECT: Draft Final Report 12-30 on *Data Center Operations*, Systemwide

In response to your memorandum of September 4, 2013, I accept the response as submitted with the draft final report on *Data Center Operations*, Systemwide.

TPW/amd

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